GUTECH, AL-GHAZALI UNIVERSITY

Department of Computer Science

MIDTERM EXAMINATION

TAKE-HOME DESIGN and IMPLEMENTATION SPRINT

Digital Well-Being and Mental Health Challenge

Course	Design Thinking	**Instructor**	Dr. Rauf Ahmed Shams Malick
Program / Semester	First Semester, Fall 26	**Total marks**	100 (plus up to 5 bonus)
Exam type	Midterm, take-home sprint	**Submission date**	_____
Team members	_____		
Team group	Girls' team / Boys' team (circle one)		

1. Course Learning Objectives assessed

This paper tests three core thinking skills: analytical, critical and creative. A fourth objective covers communication and teamwork. Each question below is tagged with the CLOs it assesses.

CLO	Skill	What you must show in this exam
CLO1	**Analytical**	Break down interviews, survey data and published reports; rank problems by evidence; trace symptoms to their root causes.
CLO2	**Critical**	Question assumptions, judge the quality and bias of evidence, weigh trade-offs, and apply ethics and safety to every design choice.
CLO3	**Creative**	Generate a wide range of original ideas, combine them into a holistic concept, and build and test a working prototype.
CLO4	**Communication and teamwork**	Work in a team, document the process honestly, and present findings clearly through a report, video and blogs.

2. Instructions to candidates

- **Work in teams of 2 to 3 students of the same gender.** Girls' teams work on problems faced by girls; boys' teams work on problems faced by boys.
- **Find the real problem through interviews, do not assume it.** The best place is the students' own setting, such as hostels or boarding places (get permission from the hostel administration, go in pairs, visit in daytime, use common areas). Interviews on campus are also accepted.
- **Minimum evidence:** 8 interviews (at least half in hostels or boarding places if access is granted), a survey with 30 responses, and 3 published surveys or reports, cited with source and year.
- **Protect people.** Interview only your own gender, get consent first, anonymise names, photos and recordings, and never collect or share explicit or private content. You are designers, not counselors.
- **If anyone mentions self-harm or seems in serious distress,** stop the interview, stay kind, do not promise secrecy, connect them with the campus counselor or a trusted faculty member on priority.
- **Declare any use of AI tools** and say where and how you used them. Marks reward evidence and reasoning, not polish alone.

3. The problem

Mobile and digital gadgets can harm student mental health in many linked ways. Reported problems include the scrolling trap, digital harassment, sharing and consuming inappropriate content, peer

pressure and social comparison from constantly watching others' success and leisure trips, living in a fantasy world, game addiction, and loneliness or misery that can include suicidal thoughts.

Your task: find which of these problems matter most to your group, prove it with evidence, dig down to the root causes, and design and build a **holistic web or mobile app** that addresses those causes rather than a single symptom. The right question leads to the right design, and the deeper you dig, the closer you get to the true cause. This is a sprint, so work through every stage of design thinking (empathize, define, ideate, prototype, test) with rigour and record what you do at each step.

4. Questions

Answer all questions. Your final report should have one section per question.

Q1. Strategy and planning 15 marks | CLO1, CLO2, CLO4

(a) Write your sprint plan: team roles, a timeline across all stages, and your evidence targets. (5)

(b) State where, whom and how you will interview, and justify these choices. (4)

(c) Set out your consent, privacy and safety steps for sensitive topics, and what your team will do if a participant shows distress. (6)

Q2. In-depth analysis 25 marks | CLO1, CLO2

(a) Collect and present your evidence: interview log, survey results and at least three cited published reports. (8)

(b) Build an evidence table that ranks the problems by how many interviews and survey responses support them, how severe they are, and whether published reports agree. Explain your ranking. (7)

(c) Dig to the root causes of your top problems using 5 Whys or a problem tree. Write a point-of-view statement and three How-might-we questions. (6)

(d) Critically assess your own evidence: possible bias, gaps, and what you cannot claim. (4)

Q3. Following the process: ideate, test and iterate 20 marks | CLO3, CLO2, CLO4

(a) Generate at least ten ideas, including some that tackle several root causes at once, and group them into one holistic concept. (6)

(b) Choose your features using clear criteria (desirability, feasibility, viability, safety and ethics) and justify what you cut. (5)

(c) Test with at least five students from your group. Record what you observed and keep a change log of what you improved. (6)

(d) Show that every stage was completed in order, with evidence of looping back when findings changed your thinking. (3)

Q4. Design and implementation 25 marks | CLO3, CLO1, CLO2

(a) Build a working web or mobile app. Provide demo access (link or install file) and a short user guide. (10)

(b) Provide a feature-to-evidence table that links each feature to a root cause and the evidence behind it. (6)

(c) Show that the app covers a broad range of your ranked problems, not only one symptom such as screen time. (5)

(d) Show safety by design: no identifiable mental-health data without consent, no monitoring of individuals by staff or parents, no shaming, and a clear path to a real person for help. (4)

Q5. Conclusions and sharing 15 marks | CLO2, CLO4

(a) Submit a final report (PDF) with one section per stage, your conclusions and lessons, and the limits of your findings. (6)

(b) Submit a final video of 3 to 5 minutes showing the problem, your evidence, the app in use and what you learned. Upload it to Google Drive and share the link on the classroom with viewing access for the class. (4)

(c) Submit a journey interview (up to 2 pages, question-and-answer form) covering what you believed at the start, what the evidence changed, your biggest pivot, what surprised you, what you would do next, and what you learned as designers. (5)

Bonus (up to 5 marks): publish at least one blog or report during the sprint and one at the end, using anonymised insights only.

5. Marks distribution

Question	Marks	What is assessed	CLOs
Q1 Strategy and planning	15	Plan, interview strategy, ethics and safety	CLO1, CLO2, CLO4
Q2 In-depth analysis	25	Evidence, prioritization, root causes	CLO1, CLO2
Q3 Following the process	20	Ideate, test and iterate with rigor, at least 3 ideas	CLO3, CLO2, CLO4
Q4 Design and implementation	25	Your individual/independent design, your agreed upon solution and working app with right features	CLO3, CLO1, CLO2
Q5 Conclusions and sharing	15	Report, video, journey interview	CLO2, CLO4
Total	**100**		
Bonus: published blogs	up to 5	Published, anonymized, useful to readers	CLO4